

LINKED LIST

Problem – 206. Reverse Linked List

Easy

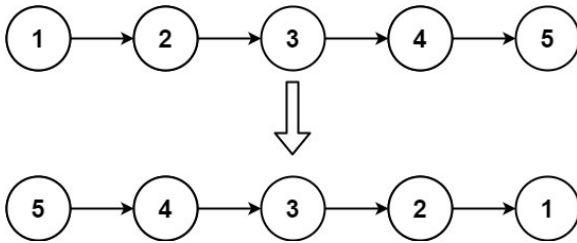


LeetCode

leetcode.com/problems/reverse-linked-list

Problem

- This is a classic problem
- Given a singly linked list, reverse its order



Solution – 206. Reverse Linked List

Easy



LeetCode

leetcode.com/problems/reverse-linked-list

Solution

- Use recursive approach
- Looking at the pseudo-code, this recursion will return the last node:

```
reverseList(head) {  
    if (!head->next) return head  
    node = reverseList(head->next);  
    return node  
}
```

- From end to beginning, each head will be a node in the list
- Therefore, you can change this node by setting a new head:

```
head->next->next = head;  
head->next = nullptr;
```

Code – 206. Reverse Linked List

Easy



LeetCode

leetcode.com/problems/reverse-linked-list

Code Time: **$O(n)$** Space: **$O(1)$**

```
ListNode* reverseList(ListNode* head) {  
    if (!head->next) return head;  
    ListNode* node = reverseList(head->next);  
    head->next->next = head;  
    head->next = nullptr;  
    return node;  
}
```

Problem – 141. Linked List Cycle

Easy



LeetCode

leetcode.com/problems/linked-list-cycle

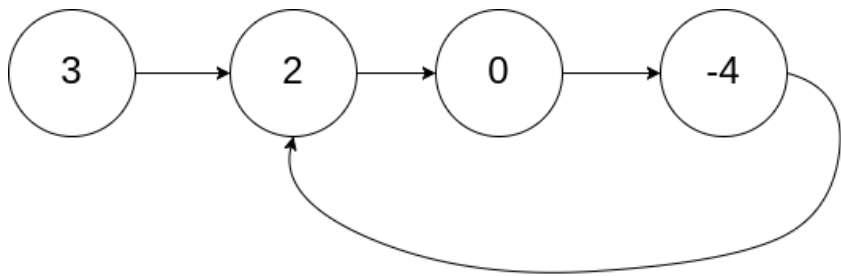
Problem

- You are given the head of a linked list
- Return **true** if there is a cycle, false otherwise

- **Example:**

In the image below, there is a cycle (-4 to 2)

Output: true



Solution – 141. Linked List Cycle

Easy



LeetCode

leetcode.com/problems/linked-list-cycle

Solution

- Have two pointers: fast and slow
- Slow will go over each item in the linked list
- Fast will go twice as fast as slow (`fast = fast->next->next`)
- If fast reach at the end, there is no cycle
- If fast encounter slow, there is a cycle, return true

Code – 141. Linked List Cycle

Easy



LeetCode

leetcode.com/problems/linked-list-cycle

Code Time: **O(n)** Space: **O(1)**

```
bool hasCycle(ListNode *head) {
    if (!head || !head->next) return false;
    ListNode* slow = head;
    ListNode* fast = head;
    while (fast && fast->next) {
        slow = slow->next;
        fast = fast->next->next;
        if (slow == fast) return true;
    }
    return false;
}
```

Problem – 21. Merge Two Sorted Lists

Easy

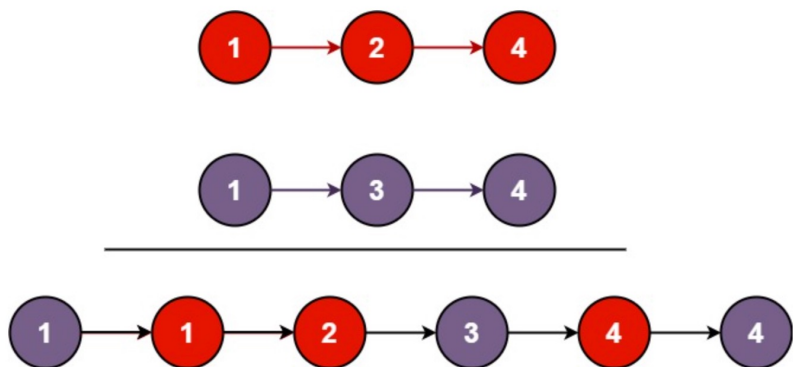


LeetCode

leetcode.com/problems/merge-two-sorted-lists

Problem

- You are given the head of two linked lists (list1 and list2)
- Merge the two lists into one **sorted** list



Solution – 21. Merge Two Sorted Lists

Easy



LeetCode

leetcode.com/problems/merge-two-sorted-lists

Solution

- Recursively explore the two lists. Base case:

```
if (!list1) return list2;
```

```
if (!list2) return list1;
```

- Compare the value of the current node of list 1 and list 2

```
if (list1->val > list2->val) { ...
```

- Set the next node of the node with the minimum value:

assume the previous condition is true, so

```
list2->next = mergeTwoLists(list1, list2->next);
```

```
return list2;
```

meaning, we want list2->next to come before list1. But we do this recursively since we need the next result

Code – 21. Merge Two Sorted Lists

Easy



LeetCode

leetcode.com/problems/merge-two-sorted-lists

Code Time: $O(n + m)$ Space: $O(n + m)$ where n is the length of list1 and m is the length of list2

```
ListNode* mergeTwoLists(ListNode* list1, ListNode* list2) {  
    if (!list1) return list2;  
    if (!list2) return list1;  
  
    if (list1->val < list2->val) {  
        list1->next = mergeTwoLists(list1->next, list2);  
        return list1;  
    } else {  
        list2->next = mergeTwoLists(list2->next, list1);  
        return list2;  
    }  
}
```

Problem – 23. Merge k Sorted Lists

Hard



LeetCode

leetcode.com/problems/merge-k-sorted-lists

Problem

- You are given an **array of k linked lists**
- Each linked list is **sorted** in ascending order
- Merge all linked lists into one **sorted** linked-lists

Solution – 23. Merge k Sorted Lists

Hard



LeetCode

leetcode.com/problems/merge-k-sorted-lists

Solution

- Create a function to merge two lists
- Go over the lists and merge with each over; **or**
- Use divide and conquer to merge (more optimal)
- Divide and conquer is more efficient because it avoids merging a big list with a small one multiple times

Code – 23. Merge k Sorted Lists

Hard



LeetCode

leetcode.com/problems/merge-k-sorted-lists

Code Time: $O(N \log k)$ Space: $O(\log k)$ where N is the total number of nodes across all lists and k is the number of lists

```
ListNode* mergeKLists(vector<ListNode*>& lists) {
    if (lists.empty()) return nullptr;
    return divideAndConquer(lists, 0 /* left */, lists.size() - 1 /* right */);
}

ListNode* mergeTwoLists(ListNode* l1, ListNode* l2) {
    if (!l1) return l2;
    if (!l2) return l1;

    if (l1->val < l2->val) {
        l1->next = mergeTwoLists(l1->next, l2);
        return l1;
    } else {
        l2->next = mergeTwoLists(l2->next, l1);
        return l2;
    }
}

ListNode* divideAndConquer(vector<ListNode*> lists, int left, int right) {
    if (left == right) return lists[right];

    int mid = left + (right - left) / 2;
    ListNode* l1 = divideAndConquer(lists, left, mid);
    ListNode* l2 = divideAndConquer(lists, mid + 1, right);
    return mergeTwoLists(l1, l2);
}
```

Problem – 146. LRU cache

Medium



LeetCode

leetcode.com/problems/lru-cache

Problem

- **LRU (Least Recently Used)** is a cache eviction policy where when a cache is full and a new item needs to be added, the least accessed item is removed.
- When an element is accessed, the item is marked as recent accessed
- More information: https://en.wikipedia.org/wiki/Cache_replacement_policies#LRU
- You are asked to design a LRU data structure
- The data structure has a given capacity
- You must implement **get** to retrieve a cache value
- You must implement **put** to add or update a new value
- Both **get** and **put** should run in **O(1)** average time complexity

Problem – 146. LRU cache

Medium



LeetCode

leetcode.com/problems/lru-cache

Solution

- Store the values in a data structure and keep the least accessed at the end
- You have to store your cache in some data structure that is **update-friendly**
- **Doubly linked** list is a good candidate. However, accessing an element costs **O(n)**
- To solve this issue, you can create a **hash map to index** each key from the doubly linked list
- You end up with two main data structures:

```
std::list<pair<int, int>> data;  
std::unordered_map<int, std::list<pair<int, int>>::iterator> index;
```

- **data** is where the key values are stored
- **index** contains the key and the iterator pointer to the data element
- Invalid capacity handling is not implemented (capacity <= 0)

Problem – 146. LRU cache

Medium



LeetCode

leetcode.com/problems/lru-cache

Code Time: **O(1)** Space: **O(capacity)**

```
class LRUCache {
private:
    std::list<std::pair<int, int>> data;
    std::unordered_map<int, std::list<std::pair<int,
int>>::iterator> index;
    int capacity;

    void moveToFront(std::list<std::pair<int,
int>>::iterator dataIt) {
        data.splice(data.begin(), data, dataIt);
    }

public:
    LRUCache(int capacity) : capacity(capacity) {}

    int get(int key) {
        auto it = index.find(key);
        if (it != index.end()) {
            auto dataIt = it->second;
            moveToFront(dataIt);
            return dataIt->second;
        }
        return -1;
    }
};
```

```
void put(int key, int value) {
    auto it = index.find(key);
    if (it != index.end()) {
        // key exists
        auto dataIt = it->second;
        dataIt->second = value;
        moveToFront(dataIt);
        return;
    }

    // key doesn't exist: check capacity and add new element
    if (data.size() >= capacity) {
        // evict least recently used (back element)
        auto last = data.back();
        index.erase(last.first);
        data.pop_back();
    }

    // add new element to front
    data.push_front({key, value});
    index[key] = data.begin();
};
```